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A PROJECT REPORT

on

“Swaraaha: Stutter Classification and Localization System”

Submitted in partial fulfilment of the requirements for the award of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING
VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt., of Karnataka

Nehru Nagar, Puttur - 574 203, DK, Karnataka, India.

November, 2026

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CERTIFICATE

Certified that the project work entitled “**Swaraaha: Stutter Classification and Localization System**” is carried out by **Mr. K Shreekrishna Upadhyaya, Mr. M Chethan Keshav Bhat, Mr. Skanda Prasad K, Mr. Srinivas Hegde M** bearing USNs **4VP23AI020, 4VP23AI023, 4VP23AI051 and 4VP23AI054** respectively bonafide students of **Vivekananda College of Engineering & Technology, Puttur** in partial fulfilment for the award of **Bachelor of Engineering in Artificial Intelligence & Machine Learning** of the **Visvesvaraya Technological University, Belagavi** during the year 2026-27. It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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DECLARATION

We, **Mr. K Shreekrishna Upadhyaya (4VP23AI020), Mr. M Chethan Keshav Bhat (4VP23AI023), Mr. Skanda Prasad K (4VP23AI051), Mr. Srinivas Hegde M (4VP23AI054)** students of B.E. 7th Semester in Artificial Intelligence & Machine Learning, **Vivekananda College of Engineering & Technology**, Puttur, hereby declare that the project work entitled “**Swaraaha: Stutter Classification and Localization System**” has been carried out by us at VCET, Puttur, under the guidance of **Prof. Ajay Shastry C G**, Assistant Professor, Department of Artificial Intelligence & Machine Learning, Vivekananda College of Engineering & Technology, Puttur, and submitted in partial fulfilment of the requirements for the award of degree in **Bachelor of Engineering in Artificial Intelligence & Machine Learning** by **Visvesvaraya Technological University**, Belagavi during the academic year 2026-2027.

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ACKNOWLEDGEMENT

We take this opportunity to express our deep heartfelt gratitude to all those people who have helped us in the successful completion of the project.

First and foremost, we would like to express our sincere gratitude to our guide, **Prof. Ajay Shastry C G.** for providing excellent guidance, encouragement and inspiration throughout the project work. Without his invaluable guidance, this work would never have been a successful one.

We would like to express our sincere gratitude to Head of the Department of Artificial Intelligence & Machine Learning, **Dr. Radhika Shetty D S** for her guidance and inspiration.

We would like to thank our Principal, **Dr. Mahesh Prasanna K.** for providing all the facilities and a proper environment to work in the college campus.

We would like to thank our management for providing the necessary infrastructure to carry out the project work.

We are thankful to all the teaching and non-teaching staff members of Artificial Intelligence & Machine Learning Department for their help and needed support rendered throughout the project.

ABSTRACT

Stuttering is a speech disorder that disrupts the natural flow of communication through involuntary repetitions, prolongations, blocks, and pauses in speech. Nearly 1% of people worldwide are impacted, and it can significantly affect social, professional, and personal life. Traditional diagnosis methods rely on manual assessment by speech-language pathologists, which are subjective, time-consuming, and not easily accessible to all individuals. To overcome these limitations, this project presents an intelligent and automated system that leverages deep learning to detect, classify, and analyze types of stuttering events in speech recordings. The system processes speech input to identify different types of dysfluencies and provides structured feedback on their occurrence. Through a simple and user-friendly interface, users can record or upload audio and receive analysis results in an accessible format. By combining speech processing and machine learning techniques, the system offers a practical and reliable tool for early stutter detection, objective speech evaluation, and ongoing therapy support, benefiting both speech-language pathologists and individuals undergoing treatment.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ASR	Automatic Speech Recognition
AUROC	Area Under the Receiver Operating Characteristic Curve
AUPRC	Area Under the Precision-Recall Curve
BERT	Bidirectional Encoder Representations from Transformers
Bi-LSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CTC	Connectionist Temporal Classification
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HMM	Hidden Markov Model
HuBERT	Hidden-Unit BERT
isWER	Intended Speech Word Error Rate
LLM	Large Language Model
LPC	Linear Predictive Coding
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstral Coefficients
PCA	Principal Component Analysis
PWS	People Who Stutter
RAG	Retrieval-Augmented Generation
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
Sep-28k	Stuttering Event Prediction Dataset (28,000 samples)
SLP	Speech-Language Pathologist
SOTA	State of the Art
SVM	Support Vector Machine
TTS	Text-to-Speech
UAR	Unweighted Average Recall
UI	User Interface
VITS	Variational Inference with Adversarial Learning for end-to-end Text-to-Speech
WavLM	Wave Language Model
WER	Word Error Rate
YOLO	You Only Look Once

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO THE PROJECT

Speech fluency is an essential aspect of human communication, allowing individuals to express thoughts and emotions clearly. However, many people face interruptions in their speech flow due to a disorder known as stuttering. It is a common speech disorder characterized by involuntary repetitions, prolongations, or blocks during speech production. This condition affects individuals across all age groups and can significantly impact their confidence, communication ability, and social interactions. Early identification of stuttering patterns is crucial for effective therapy and management.

Traditionally, detection and assessment of stuttering rely on manual observation by speech-language pathologists, which can be subjective, time-consuming, and often inconsistent across experts, causing physical fatigue. Such dependence on human evaluation makes large-scale assessment and continuous monitoring difficult. With recent advancements in artificial intelligence, especially in the field of deep learning and speech signal processing, automated stutter detection has become a reliable and scalable alternative. The project aims to develop an intelligent system that can automatically detect, classify, and analyze types of stuttering in speech, including repetitions, prolongations, blocks, and fillers, using deep learning techniques.

The input audio will be first preprocessed and converted into embeddings, which will be then analyzed using Wav2Vec 2.0 to learn both local acoustic patterns and long-range temporal dependencies that differentiate normal and dysfluent speech. Along with stutter classification, the same audio will be also passed through a localization module, to identify timestamps of the stuttering event. The final model will be integrated into a user-friendly interface that allows users to record or upload speech samples and receive analysis, timestamped detection results, and classification output.

Through this approach, the project seeks to support speech-language pathologists and individuals affected by stuttering by providing an accurate, objective, and accessible diagnostic tool. It also contributes to the broader vision of integrating artificial intelligence into healthcare to enhance diagnosis, therapy, and quality of life.

1.2 LITERATURE REVIEW

P. Arbajian et al. [1] have proposed a speech segment selection strategy for improving stutter block detection accuracy. Their approach involves extracting acoustic features from annotated speech samples and employing machine learning classifiers to evaluate performance under different segmentation setups. They studied the effect of temporal segmentation on the detection of stuttering events by comparing different windowing approaches. The results indicate that segment choice has a strong impact on accuracy - some segment lengths are better at modeling dysfluency patterns, while bad segmentation can lose important temporal information and lead to degraded performance. The authors emphasize that proper speech segmentation improves detection accuracy and the effectiveness of remediation systems. This finding is directly pertinent to the proposed system, as it highlights the importance of preprocessing, in particular optimized speech segmentation, to enhance the performance of real-time stutter detection models.

V. Mitra et al. [2] have developed a tuning-based approach to enhance voice assistant performance for dysfluent speech by optimizing an existing hybrid ASR system. The goal is to reduce recognition errors due to stuttering, such as repetitions and unintended insertions. The methodology is based on tuning the important decoding parameters of the ASR system - more weight is given to the language model by increasing the penalty for inserting words and decreasing the weight of the acoustic model. This change helps the system to filter out repeated or disfluent segments better and focus on meaningful speech patterns. The system was tested on speech data from 18 participants with varying degrees of stuttering severity, achieving a 24% relative reduction in intended speech Word Error Rate (isWER). The authors point out that dysfluencies can be successfully addressed by tuning parameters in ASR systems without major architectural changes. This work supports the proposed system in underscoring the importance of model tuning and post-processing for stuttered speech.

P. Mohapatra et al. [3] have introduced disfluencyNet, a deep learning model for automatic detection of speech disfluencies using contextual representations. The model employs contextual embeddings to better model speech dependencies and enhance the identification of disfluencies such as repetitions and pauses. Their approach provides contextual embeddings to a classification network and obviates the need for large training datasets through data distillation techniques. The model was trained and evaluated on benchmark datasets such as SEP-28k and FluencyBank, achieving competitive accuracy

against baseline models using only a quarter of the data. This demonstrates the efficiency and strong generalization ability of the model. The authors note that contextual representations and data-efficient training techniques can significantly reduce the need for large annotated datasets. This work is pertinent to the proposed system, as it validates the use of contextual embeddings and efficient training strategies to address data scarcity and bolster robustness in real-time stutter detection systems.

J. Liu et al. [4] have introduced a novel method for detecting disfluencies in speech utilizing the context-based embeddings provided by wav2vec 2.0 for dealing with speech data from different languages and differing speech lengths. The proposed solution includes a classification neural network enhanced with wav2vec 2.0 embeddings and employs data distillation to select high-quality audio fragments where three human annotators agree on disfluency labels. The evaluation was conducted on multilingual datasets characterized by speech of different lengths, demonstrating superior performance compared to other approaches. The authors report that combining pretrained models with high-quality distilled data significantly boosts detection accuracy and reduces noise in training. These findings are directly applicable to the proposed system, as they reinforce the use of pretrained models like wav2vec 2.0 and data filtering techniques to enhance robustness, efficiency, and cross-speaker generalization in real-time stutter detection systems.

A. Romana et al. [5] have presented a multimodal approach for detecting speech disfluencies directly from untranscribed audio. The model combines both acoustic and linguistic information to improve detection accuracy without relying on perfect transcriptions. The methodology uses a Bi-LSTM-based fusion model that operates at the frame level, integrating WavLM acoustic features with BERT-based language representations derived from transcripts generated by a fine-tuned Whisper model. This combination helps the system capture both low-level speech patterns and high-level contextual information, effectively addressing common issues such as ASR transcription errors and misalignment. The results show improved robustness and accuracy in detecting disfluencies compared to single-modality approaches, especially in real-world noisy conditions. This work is directly pertinent to the proposed system, as it validates integrating acoustic and language features to improve detection accuracy and robustness in real-time stutter detection systems.

D. Wagner et al. [6] have presented a hybrid approach that combines acoustic and linguistic representations using Large Language Models (LLMs) for dysfluency detection.

The system captures both speech patterns and textual context to improve classification performance. The methodology involves extracting acoustic features using wav2vec 2.0 and generating transcriptions through Whisper ASR, which are then fused into a joint input and processed by an LLM to classify different stutter types including repetitions, prolongations, and blocks. The results show that this combined acoustic-lexical approach outperforms models that rely solely on either audio or text features, demonstrating improved accuracy and robustness. The authors observe that integrating multimodal inputs with LLMs enhances the model's ability to understand complex speech patterns and contextual cues. This study is directly pertinent to the proposed system, as it reinforces the value of multimodal fusion and advanced models like LLMs to improve accuracy and generalization in real-time stutter detection systems.

S. A. Sheikh et al. [7] have developed a deep learning-based framework called multi-contextual (MC) StutterNet for improving stuttering detection by addressing key challenges such as class imbalance and limited data availability. The methodology incorporates a multi-branching architecture that processes different contextual representations of speech. To handle class imbalance, the model applies class-balanced loss by assigning appropriate weights to underrepresented stutter categories, while data augmentation techniques are used to increase dataset diversity and improve generalization. The results demonstrate improved robustness and accuracy in detecting stuttering events across different speech conditions, especially for less frequent dysfluency types. The authors observe that combining data augmentation, balanced loss functions, and multi-context learning significantly enhances model performance. These findings are directly applicable to the proposed system, as they inform strategies for handling data imbalance and bolstering robustness using advanced deep learning approaches for real-time stutter detection.

X. Zhou et al. [8] have designed a YOLO-inspired deep learning model for speech dysfluency detection by treating spectrograms as images, enabling simultaneous localization and classification of dysfluencies within the time–frequency domain. The methodology involves extracting spectrograms from speech and applying a region-based detection model that learns both spatial features (frequency patterns) and temporal dynamics (changes over time). Additionally, speech-text alignment is incorporated to associate audio segments with corresponding words, improving contextual understanding. The model predicts both the type of dysfluency and the exact time region where it occurs, enabling precise and interpretable detection. The results demonstrate strong performance in identifying multiple

dysfluency types with accurate localization. This approach is directly applicable to the proposed system, as it supports real-time, region-wise stutter detection using spectrogram-based CNN architectures.

X. Zhou et al. [9] have introduced stutter-Solver, a YOLO-inspired end-to-end model designed for multilingual dysfluency detection that identifies both the type and temporal location of stuttering events across different languages. The methodology involves treating speech spectrograms as image-like inputs and applying a region-based detection framework. To address data scarcity, the authors generate synthetic datasets such as VCTK-Pro, VCTK-Art, and AISHELL3-Pro using articulatory and text-to-speech (TTS) based simulations, enhancing data diversity and supporting multilingual learning. The results demonstrate state-of-the-art performance across multiple datasets and languages, highlighting the effectiveness of synthetic data augmentation and region-based detection. These findings are directly applicable to the proposed system, as they support multilingual capability, synthetic data usage, and real-time region-wise detection for robust stutter detection systems.

J. Zhang et al. [10] have introduced LLM-Dys, a novel methodology for generating large-scale dysfluent speech datasets using Large Language Models (LLMs) to overcome data scarcity and improve diversity in dysfluency detection tasks. The methodology involves using an LLM to simulate realistic dysfluency patterns and generate labeled dysfluent text, which is then converted into speech using the VITS model, producing high-quality synthetic audio. Unlike traditional rule-based methods, this approach captures more natural prosody and contextual variation. The dataset includes 11 categories of dysfluencies at both word and phoneme levels, enabling fine-grained analysis and model training. The results demonstrate improved data quality and diversity, leading to better model performance in detection tasks. This work is pertinent to the proposed system, as it advances data augmentation using LLMs and TTS, bolstering robustness and accuracy in stutter detection systems.

S. Kim and A. Kumar [11] have developed fluentNet, a hybrid CNN-LSTM architecture designed to automatically capture both spatial and temporal dependencies in speech signals for end-to-end detection of speech disfluency. The CNN layers extract short-term spectral features from Mel spectrograms, while the LSTM layers model temporal continuity, making it effective in identifying recurring stutter patterns over time. The system

was trained and validated on the SEP-28k dataset, achieving over 91% classification accuracy. The paper highlights the advantage of end-to-end models that do not rely on handcrafted features, thus reducing bias and improving generalization across speakers and environments. The authors also demonstrated FluentNet's real-time applicability in speech therapy by integrating it into a feedback loop that provides visual dysfluency indicators to users. This work directly supports the proposed system, as it validates the effectiveness of CNN-LSTM architectures for real-time audio-based classification and informs the multi-model CNN approach used in our project.

R. Ahmed and J. Park [12] have developed a multilingual speech dysfluency detection framework capable of identifying stuttering events across English, Korean, and Japanese datasets. The model employs an encoder-decoder transformer architecture similar to BERT, enabling it to capture contextual relationships in speech sequences. It was trained on combined multilingual datasets, showing high adaptability and robustness to linguistic variations. The model achieved an F1-score of 95.2% in English and over 93% in non-English corpora. The study underscores the importance of language-independent feature representations for building universally accessible stutter detection systems. The authors also incorporated explainable AI techniques to visualize attention weights, showing which segments of the input contributed most to the detection decision. This research provides valuable insight for the current project, particularly in the areas of cross-language model generalization and interpretability in dysfluency detection systems.

F. Rahimi and D. Torres [13] have presented an exploration of large language models (LLMs) and transformer-based architectures for speech dysfluency analysis. By embedding speech features as tokenized sequences and applying self-attention mechanisms, the model effectively detects contextual disruptions in spoken language patterns. The authors compared their LLM-based approach with traditional CNN and RNN models, finding that transformers outperformed them in both recall and precision, particularly for subtle dysfluency types like interjections and soft blocks. The study also shows that pretraining on large general speech datasets improves performance even on smaller stutter-specific corpora. The integration of explainability tools such as attention visualization provides transparency in predictions. This work reinforces the growing relevance of transformer-based models and encourages future exploration into combining CNN-based acoustic analysis with language-level contextual understanding for enhanced stutter detection accuracy.

V. Uloza et al. [14] have conducted a study on AI applications for analyzing pathological speech characteristics, particularly substitution voicing, using deep neural networks. The study applies CNNs and principal component analysis (PCA) to classify voice disorders based on acoustic features. The authors stress the importance of preprocessing techniques such as noise removal and normalization for ensuring consistent input to neural networks. The results showed a classification accuracy exceeding 93%, validating the capability of AI to detect subtle speech impairments. Though the focus is on substitution voicing rather than stuttering, the methodology provides essential insights into building speech pathology systems that rely on spectral analysis. In the context of the proposed stuttering detection system, this research supports the use of Mel spectrograms and CNN-based feature extraction for effective dysfluency recognition.

H. Müller and C. Lee [15] have analyzed the neural bases involved in speech production of stutterers through a meta-analysis of fMRI and EEG studies. The authors identify specific neural regions such as the inferior frontal gyrus and basal ganglia that exhibit atypical activation during speech production. These findings provide biological validation for AI-based stutter detection systems, which often rely on acoustic signatures reflecting underlying neural control differences. While the study does not propose a computational model, it offers theoretical grounding that explains why stuttering manifests in measurable acoustic patterns. For this project, the insights are crucial for feature selection and interpretation, linking speech irregularities detected by the model to their neurological causes.

A. Baevski et al. [16] have introduced a self-supervised framework for learning speech representations from raw audio waveforms. The model uses a convolutional feature encoder to extract latent representations, followed by a transformer encoder that contextualizes them through masked prediction. Pre-trained on 960 hours of unlabeled LibriSpeech data and fine-tuned with a CTC loss for downstream tasks. Achieved state-of-the-art results on phoneme recognition, speaker identification, and emotion recognition benchmarks. This work provides the foundational speech representation model used in the proposed system, enabling robust feature extraction from stuttered speech without requiring large labeled datasets.

P. Khanna et al. [17] have developed a multimodal stuttering detection system combining acoustic features from Wav2Vec 2.0 and HuBERT with linguistic features from Llama-2, enhanced by Retrieval-Augmented Generation for adaptive classification. The system

fuses acoustic and linguistic embeddings through a fusion strategy that weights modalities based on input quality. Evaluated on SEP-28k and FluencyBank datasets, achieving state-of-the-art performance across all stuttering event types. The RAG component retrieves similar historical cases to improve classification of rare stuttering patterns. This multi-modal fusion approach with adaptive retrieval directly supports the proposed system's goal of combining audio and text features for comprehensive stutter analysis.

C. Lea and V. Mitra [18] have presented a large-scale dataset of over 28,000 speech clips extracted from stuttering support group podcasts, annotated with five stuttering event types: blocks, prolongations, repetitions, interjections, and revisions. The dataset includes both crowdsourced annotations from non-expert listeners and expert annotations from speech-language pathologists. Experiments show that increasing annotation scale from 10k to 28k clips yields 28% and 24% F1 improvements for blocks and prolongations respectively. This dataset addresses the critical data scarcity problem in stuttering research and provides the primary benchmark for evaluating the proposed detection system.

O. Shonibare et al. [19] have proposed a two-stage approach for improving automatic speech recognition on stuttered speech with limited labeled data. The first stage uses a context-aware classifier trained on small amounts of labeled stuttering data to detect dysfluency regions. The second stage modifies the ASR decoder to pass over detected dysfluencies, focusing transcription on fluent segments. Achieved 12-71% word error rate reduction across different stuttering severity levels. This detect-then-transcribe paradigm demonstrates that dysfluency detection and speech processing can be decoupled effectively, supporting the proposed system's modular architecture.

S. Bayerl et al. [20] have designed application of fine-tuned wav2vec 2.0 for detecting dysfluencies in clinical stuttering therapy recordings. The system combines wav2vec 2.0 acoustic embeddings with a multi-task learning framework and SVM classifier to distinguish between different dysfluency types in therapy speech. Evaluated on FluencyBank and the German KSoF dataset, achieving 27% F1 improvement over baseline methods. The multi-task approach simultaneously predicts dysfluency presence and type, enabling therapists to track progress over sessions. This clinical application validates the use of pre-trained speech models for real-world stuttering assessment.

S. Bayerl et al. [21] have introduced a modified wav2vec 2.0 framework that treats stuttering detection as a multi-label classification problem, recognizing that dysfluencies frequently co-occur in natural speech. The model assigns independent probability scores to each dysfluency type, enabling simultaneous detection of multiple overlapping events. Achieved state-of-the-art results on the SEP-28k-Extended dataset and demonstrated cross-language generalization from English to German. This multi-label formulation better captures the reality of stuttering speech where a single utterance may contain blocks, prolongations, and repetitions simultaneously.

R. Gong et al. [22] have presented the first and largest Mandarin Chinese stuttered speech dataset, containing 70 hours of recordings from speakers with varying stuttering severity levels. The dataset includes verbatim transcriptions with detailed dysfluency annotations at word and syllable levels. Covers diverse speaker demographics including different age groups, genders, and stuttering types. Released as an open-source resource to encourage Mandarin stuttering research. This dataset expands the linguistic coverage of stuttering detection beyond English and enables development of multilingual systems.

X. Liu et al. [23] have designed an end-to-end stuttering detection architecture combining Conformer blocks for local acoustic pattern extraction with BiLSTM layers for long-range temporal dependency modeling. The multi-task framework simultaneously predicts dysfluency types and severity levels. Achieved first place in the SLT 2024 Stuttering Detection Challenge and 39.8% F1 improvement on the AS-70 Mandarin dataset. The Conformer's ability to capture both local and global speech patterns through combined convolution and self-attention mechanisms proves particularly effective for detecting the varied temporal signatures of different stuttering events.

A. Batra et al. [24] have introduced a multi-lingual Indian language dataset for understanding stuttering experiences, containing both read and spontaneous speech across five stuttering types. The dataset captures real-world stuttering patterns from speakers of Hindi, Kannada, Telugu, and other Indian languages. Includes demographic metadata and self-reported stuttering severity ratings. Addresses the significant gap in non-English stuttering datasets, particularly for South Asian languages where stuttering prevalence and manifestation patterns differ from Western populations.

A. R. Valente et al. [25] have developed an enhanced version of the FluencyBank dataset with detailed clinical annotations provided by expert speech-language pathologists. Annotations include audiovisual cues, secondary stuttering behaviors such as facial tension and eye blinking, and physiological tension indicators beyond traditional acoustic dysfluency labels. The multi-dimensional annotation scheme captures the full clinical picture of stuttering severity. These expert annotations enable training of more clinically accurate detection models that consider the complete stuttering experience rather than just acoustic events.

T. Grósz et al. [26] have analyzed a wav2vec2-based paralinguistic analysis framework that jointly recognizes vocalised emotions and stuttering events from speech. The system extracts self-supervised speech embeddings and processes them through task-specific classification heads for emotion recognition and stuttering detection. Achieved 62.1% unweighted average recall on the Stuttering Sub-Challenge benchmark. This multi-task paralinguistic approach demonstrates that emotional state and speech disfluency are interconnected, suggesting that joint modeling can improve detection accuracy through shared representation learning.

J. Tang et al. [27] have presented comprehensive speech annotation guidelines developed in collaboration with people who stutter, establishing standardized protocols for labeling stuttering events in speech datasets. The guidelines address challenges in annotating ambiguous dysfluencies, providing clear categorical definitions and decision trees for consistent labeling. Includes recommendations for annotator training, inter-annotator agreement measurement, and handling edge cases. These community-informed guidelines improve annotation quality and reproducibility across stuttering research, directly benefiting dataset construction and model evaluation.

A. Romana et al. [28] have developed an updated version of the FluencyBank dataset with word-level timestamps and refined disfluency annotations. The dataset provides precise temporal alignment for speech segments, enabling more accurate analysis of how speech processing models handle disfluent input. Includes updated transcripts with more granular disfluency labels and word timing information for each speech clip. This resource addresses the need for temporally precise annotations in stuttering research, supporting development of models that can localize dysfluencies at the word level.

L. Nie et al. [29] have designed the first multi-modal neural framework for stuttering detection that combines audio and visual signals through transformer-based cross-modal fusion. The model processes speech embeddings alongside facial expression and lip movement features to capture both acoustic and visual manifestations of stuttering. Achieved 2-17% F1-score improvement over existing state-of-the-art uni-modal approaches on benchmark datasets. This work demonstrates that incorporating visual information significantly aids stuttering detection, particularly for events like blocks and prolongations that have distinct visual signatures.

R. P. Buzzeti et al. [30] have proposed a two-stage hybrid approach for automatic detection and classification of stuttering-related disfluencies in Brazilian Portuguese. The first stage applies rule-based detection to identify potential disfluency regions, while the second stage uses machine learning classification for severity assessment. Addresses language-specific stuttering patterns in Portuguese, which differ from English in terms of syllable structure and dysfluency manifestation. This work extends stuttering detection to underrepresented languages and demonstrates the effectiveness of hybrid approaches that combine linguistic rules with data-driven classification.

Table 1.1: Summary of Literature Survey

Sl. No	Author	Title	Features	Pros	Cons
1	Pierre Arbajian et al.	Effect of speech segment samples selection in stutter block detection and remediation	Analyzed different speech segment lengths and sample selection methods for accurate stutter block detection using acoustic classifiers.	Improves detection precision through better segment selection.	Sensitive to segmentation configuration.
2	Vikramjit Mitra et al.	Analysis and Tuning of a Voice Assistant	Modified ASR decoding parameters by increasing	Enhances intended speech recognition for dysfluent users.	Focuses on recognition rather than dys-

Sl. No	Author	Title	Features	Pros	Cons
		System for Dys-fluent Speech	word insertion penalty and reducing acoustic model influence to suppress repetition errors		fluency classification.
3	Payal Mohapatra et al.	Speech Disfluency Detection with Contextual Representation and Data Distillation	Developed DisfluencyNet using contextual embeddings and distilled high-confidence samples for efficient low-resource training	Achieves strong results with reduced training data	Depends on carefully filtered annotations
4	Jiajun Liu et al.	Automatic Speech Disfluency Detection Using Wav2Vec 2.0 for Different Languages	Applied Wav2Vec 2.0 contextual embeddings for multilingual disfluency detection across variable speech durations.	Supports multilingual detection with strong contextual learning.	High resource requirements and limited availability of multilingual stuttering datasets.
5	Amrit Romana et al.	Automatic Disfluency Detection from Untranscribed Speech	Built multimodal Bi-LSTM combining WavLM acoustic signals with BERT linguistic features from Whisper transcripts.	Detects disfluencies without manual transcription.	Multimodal design increases system complexity.

Sl. No	Author	Title	Features	Pros	Cons
6	Dominik Wagner et al.	Large Language Models for Dysfluency Detection in Stuttered Speech	Combined Wave2Vec 2.0 audio, Whisper text, and LLM-based fusion for multi-type stutter classification.	Improves multi-class detection accuracy.	Requires heavy computational resources.
7	Shakeel A. Sheikh et al.	Advancing Stutter Detection via Data Augmentation, Class-Balanced Loss and Multi-Contextual Deep Learning	Proposed multi-contextual StutterNet with augmentation and weighted loss for robust stuttering detection.	Addresses data scarcity and class imbalance.	Synthetic augmentation may reduce realism.
8	Y. Zhang et al.	YOLO-Stutter: End-to-End Region-Wise Speech Dysfluency Detection	Adapted YOLO on speech spectrograms for simultaneous dysfluency type prediction and temporal localization.	Enables precise real-time stutter localization.	Uses small dataset; lacks multimodal or contextual input for better generalization.
9	Xuanru Zhou et al.	Stutter-Solver: End-to-End Multi-Lingual Dysfluency Detection	Developed multilingual YOLO-based dysfluency detector using synthetic articulatory and TTS-generated datasets.	Expands multilingual coverage with SOTA accuracy.	Synthetic speech may not fully match real speech.

Sl. No	Author	Title	Features	Pros	Cons
10	Jinning Zhang et al.	Analysis and Evaluation of Synthetic Data Generation in Speech Dysfluency Detection	Proposed LLM-Dys framework using LLM-generated dysfluent text and VITS TTS for scalable corpus creation.	Generates diverse large-scale dysfluency datasets.	Synthetic prosody may limit robustness.
11	S. Kim & A. Kumar	FluentNet: End-to-End Detection of Speech Disfluency with Deep Learning	CNN-LSTM hybrid model analyzing temporal and spectral dependencies in speech signals using SEP-28k dataset.	High accuracy in multi-class dysfluency detection; suitable for real-time use.	Model complexity increases training time and requires GPU-based systems.
12	R. Ahmed & J Park	Stutter-Solver: End-to-End Multi-Lingual Dysfluency Detection	Transformer-based multilingual model for detecting stuttering across multiple languages using contextual embeddings.	Supports cross-lingual generalization and explainability through attention visualization.	High resource requirements and limited availability of multilingual stuttering datasets.
13	F. Rahimi & D. Torres	Large Language Models for Dysfluency Detection in Stuttered Speech	Used transformer-based large language models (LLMs) to capture context disruption.	Expands multilingual coverage with SOTA accuracy.	Synthetic speech may not fully match real speech.

Sl. No	Author	Title	Features	Pros	Cons
			tions in stuttered speech		
14	V. Uloza et al.	An AI-Based Algorithm for the Assessment of Substitution Voicing	CNN and PCA combine feature extraction and classification of pathological voice disorders.	Demonstrates effectiveness of AI in medical speech analysis; high accuracy (>93%).	Focused on substitution voicing, not directly stuttering-related; limited dataset scope.
15	H Muller & C. Lee	Reinvestigating the Neural Bases Involved in Speech Production of Stutterers: An ALE Meta-Analysis	Analyzed fMRI/EEG studies to identify brain regions linked to speech dysfluency.	Provides a neurophysiological foundation supporting acoustic-based AI analysis.	Not a computational model; lacks implementation for automated detection.
16	A. Baevski et al.	wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations	Self-supervised pre-training of speech representations using masked prediction with transformer and convolutional encoder on raw audio.	Provides foundational speech embeddings used by many downstream stuttering detection models.	Requires large unlabeled data for pre-training; not stuttering-specific.
17	P. Khanna et al.	StuD: A Multimodal Approach for Stuttering Detection with RAG and Fusion Strategies	Combined Wav2Vec 2.0, HuBERT acoustic features with Llama-2 linguistic features and RAG-based	Achieves SOTA on SEP-28k and FluencyBank with adaptive retrieval.	Heavy computational requirements for LLM and RAG components.

Sl. No	Author	Title	Features	Pros	Cons
			adaptive classification.		
18	C. Lea and V. Mitra	SEP-28K: A Dataset for Stuttering Event Detection from Podcasts	Large-scale dataset with 28k+ clips annotated for 5 stuttering event types from podcast recordings.	Enables large-scale training with 28%/24% F1 gains from data scaling.	Podcast speech may not generalize to clinical or spontaneous settings.
19	O. Shonibare et al.	Enhancing ASR for Stuttered Speech with Limited Data using Detect and Pass	Two-stage approach: context-aware dysfluency detection followed by ASR that passes over detected events.	Achieves 12-71% WER reduction with minimal labeled data.	Relies on accurate first-stage detection; cascaded errors possible.
20	S. Bayerl et al.	Detecting Dysfluencies in Stuttering Therapy Using wav2vec 2.0	Fine-tuned wav2vec 2.0 with multi-task learning and SVM for therapy speech analysis.	27% F1 improvement on clinical therapy recordings.	Limited to therapy contexts; may not generalize to casual speech.
21	S. Bayerl et al.	Dysfluencies Seldom Come Alone - Detection as a Multi-Label Problem	Modified wav2vec 2.0 for simultaneous multi-label detection of co-occurring dysfluencies.	SOTA on SEP-28k-Extended with cross-language generalization.	Multi-label training increases model complexity and annotation requirements.
22	R. Gong et al.	AS-70: A Mandarin Stuttered	First large-scale Mandarin stut-	Enables Mandarin stuttering	Limited to Mandarin; an-

Sl. No	Author	Title	Features	Pros	Cons
		Speech Dataset for Automatic Speech Recognition	tered speech dataset with verbatim transcriptions and detailed annotations.	research with 70h of diverse speaker data.	notation process labor-intensive.
23	X. Liu et al.	An End-to-End Stuttering Detection Method Based on Conformer and BiLSTM	Conformer blocks with BiLSTM temporal modeling for multi-task dysfluency and severity prediction.	1st place SLT 2024 Challenge; 39.8% F1 improvement on AS-70.	Conformer architecture requires significant GPU memory and training time.
24	A. Batra et al.	Boli: A Dataset for Understanding Stuttering Experience	Multi-lingual Indian language dataset with read and spontaneous speech across 5 stutter types.	Captures real-world Indian stuttering patterns across multiple languages.	Limited annotations; may need expert review for clinical applications.
25	A. R. Valente et al.	Clinical Annotations for Automatic Stuttering Severity Assessment	Enhanced FluencyBank with expert clinician annotations including audio-visual and behavioral cues.	Provides clinical-grade multi-dimensional annotations for severity assessment.	Expert annotation is expensive and time-consuming to produce.
26	T. Grósz et al.	Wav2vec2-based Paralinguistic Systems to Recognise Vocalised Emotions	Joint wav2vec2 framework for emotion recognition and stuttering detection	Demonstrates emotion-disfluency interconnection with 62.1% UAR.	Joint modeling may introduce task interference; limited

Sl. No	Author	Title	Features	Pros	Cons
		tions and Stuttering	from speech embeddings.		to paralinguistic analysis.
27	J. Tang et al.	Speech Annotation Guidelines with People Who Stutter	Standardized annotation protocols developed collaboratively with PWS for consistent dysfluency labeling.	Improves annotation quality and reproducibility across stuttering research.	Guidelines adoption requires community-wide coordination and training.
28	A. Romana et al.	FluencyBank Timestamped: An Updated Data Set for Disfluency Detection and Automatic Intended Speech Recognition	Updated FluencyBank with word-level timestamps and refined disfluency annotations for precise temporal alignment.	Enables word-level dysfluency localization and more accurate model evaluation.	Limited to English stuttering patterns; requires updated annotation protocols.
29	L. Nie et al.	MMSD-Net: Towards Multimodal Stuttering Detection	First multimodal neural framework combining audio and visual signals through transformer-based cross-modal fusion.	2-17% F1 improvement over uni-modal approaches; captures visual stuttering cues.	Requires video input; increased computational complexity and data collection overhead.
30	R. P. Buzzeti et al.	Detecting Stuttering with Artificial Intelligence: A Hybrid Method for	Two-stage hybrid approach combining rule-based detection with ML classi-	Extends stuttering detection to under-represented lan-	Language-specific rules may not transfer to other languages

Sl. No	Author	Title	Features	Pros	Cons
		Brazilian Portuguese	fication for Portuguese stuttering.	guages; clinically applicable.	without modification.

1.3 EXISTING SYSTEM

Existing stutter detection systems primarily rely on manual identification or basic signal processing methods. Speech pathologists typically analyze recorded samples using auditory perception and visual cues such as waveforms or spectrograms. While experienced experts can provide valuable insights, this method suffers from inconsistencies due to human bias and limited scalability.

Some research-based tools and academic prototypes have attempted to automate stutter detection using traditional audio analysis or shallow machine learning methods like Support Vector Machines (SVM) and Hidden Markov Models (HMM). However, these systems often struggle to generalize across diverse speakers, accents, and recording environments. Moreover, they require extensive feature engineering and are not optimized for real-time or user-interactive use. Commercially available speech therapy applications, though helpful, are often subscription-based and lack specialized diagnostic features for detecting specific stutter types. Many rely on internet connectivity and do not offer offline functionality, which limits accessibility in under-resourced regions.

These limitations highlight the need for an advanced, reliable, and accessible stutter detection system. A deep learning-based approach using Wav2Vec2 embeddings, and audio spectrogram analysis can automatically learn complex acoustic patterns and temporal dependencies that distinguish normal speech from dysfluent speech, enabling accurate classification and localization of various stuttering types.

1.4 PROBLEM STATEMENT

Stuttering assessment is largely manual, subjective, and time-consuming, leading to pathologist fatigue and limited scalability. Existing automated systems can classify stuttering types but lack precise word-level localization and effective support for pathologist verification. A clinically useful solution must therefore provide objective detection with accurate word-level localization of stuttering events.

1.5 OBJECTIVES OF THE PROJECT

- **Preprocessing Pipeline:** To preprocess and standardize speech audio by reducing noise, normalizing volume, and preparing it for accurate stutter detection.
- **Classification Model:** To develop an automated stutter detection system that accurately identifies and classifies five types of stuttering: Prolongation, Block, Sound Repetition, Word Repetition, and Interjection.
- **Localization Model:** To implement a temporal localization module capable of identifying the exact timestamps of stuttering events, the affected region, and the duration within a speech sample.

CHAPTER 2

REQUIREMENT SPECIFICATION AND ANALYSIS

2.1 INTRODUCTION

The requirements analysis and specification stage of the Swaraaha: Stutter Classification and Localization System defines the main functions of the system, the expected performance, the dependencies on technical issues, the needs of users (including developers, clinicians, and end-users), etc. This phase brings all interested parties together (developing agency, clinicians, and end users) to agree on what the system will do. The ultimate goal of the system is to take speech audio recordings and use a deep learning pipeline to classify stuttering types and localize dysfluency events within the speech signal. Clearly defining at this point all functional and non-functional requirements provides a roadmap for the development process, thereby reducing the likelihood of design errors occurring and ensuring that the completed system meets the clinical and technical requirements.

2.2 FUNCTIONAL REQUIREMENTS

The functional requirements of the proposed system define the core features and operations necessary for automated stuttering detection and analysis.

2.2.1 Dataset

The system is trained and evaluated using three publicly available stuttering datasets. Project Boli provides multilingual Indian language stuttering data sourced via GitHub repositories. SEP-28K is a large-scale dataset containing approximately 28,000 audio clips annotated for five stuttering event types, obtained via Kaggle. UCLASS provides additional stuttering speech data, also obtained via Kaggle.

Each dataset is normalized into a unified format consisting of a `combined_labels.csv` file with multi-label binary annotations and corresponding interval files for each audio clip. The five dysfluency classes are prolongation, block, sound repetition, word repetition, and interjection. The complete dataset is split into training, validation, and testing subsets using an 80:10:10 ratio. An automated pipeline handles downloading, merging, and pre-processing the data to ensure efficient dataset preparation.

2.2.2 Preprocessing

Before any analysis is performed on the input audio, it must undergo a standardized preprocessing pipeline to ensure consistency and improve model performance.

The audio is first converted into a 16 kHz mono WAV format using FFmpeg, ensuring uniform input across all processing stages. DC offset is removed by subtracting the mean of the signal, which eliminates any constant bias in the waveform. Peak normalization scales the audio to a target peak amplitude of 0.95, ensuring consistent volume levels across different recordings. Silent segments are trimmed from the beginning and end of the audio using energy-based detection, reducing unnecessary padding and focusing the analysis on active speech regions.

The processed audio is then padded or truncated to a fixed length of 48,000 samples, corresponding to a duration of 3 seconds at 16 kHz. This standardization ensures that all inputs have uniform dimensions for model processing.

Data augmentation techniques are applied during training to increase dataset diversity and improve generalization. These include random noise injection, time stretching, pitch shifting, temporal shifting, and scaling. Spectrogram-level augmentations such as time masking and frequency masking are also employed to improve robustness.

2.2.3 User Requirements

The user can easily and clearly upload or record a speech sample via the system's user interface. The system must be able to receive audio files in common formats including WAV, MP3, FLAC, and M4A, and process them prior to classification.

The system will automatically classify the submitted audio, providing results to the user without requiring manual intervention during the classification process. The provided results will clearly indicate which dysfluency types are present in the speech, along with confidence scores for each detected type.

The system will localize the exact temporal positions of detected dysfluencies within the audio, presenting timestamped results that identify where each dysfluency event occurs. If the system is unable to provide a conclusive classification, it will indicate low confidence and suggest that the user consult a speech-language pathologist for further evaluation.

2.2.4 System Requirements

The system shall preprocess input audio by converting it to 16 kHz mono format, removing DC offset, applying peak normalization (up to 0.95), and trimming silence segments. For feature extraction, the system shall generate high-level speech representations using Wav2Vec 2.0 embeddings, capturing both acoustic and contextual characteristics of the input audio.

The classification pipeline shall detect five types of dysfluencies—prolongation, block, sound repetition, word repetition, and interjection—using five parallel binary classification models. The per-classifier outputs shall be aggregated into a multi-label result that reports the probability of each dysfluency type and summarizes the detected classes.

The system shall generate a timestamped transcript of the input audio using the Whisper model, supporting multiple languages including English, Kannada, and Hindi. Dysfluency localization shall be performed using CNN-based spectrogram analysis and Wav2Vec2 frame-level feature extraction, followed by alignment with corresponding words or syllables using Connectionist Temporal Classification (CTC) based time alignment.

The system shall provide visual representations of the analysis, including waveform displays with dysfluency overlays, spectrograms, and prediction probability graphs. A detailed analysis report shall be generated and maintained using local storage mechanisms.

2.3 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements define the quality attributes and operational constraints of the system. These requirements ensure that the system performs efficiently, remains user-friendly, and can be maintained and extended over time.

2.3.1 Reliability

Reliability is critical in any system used to assist clinical assessment, as incorrect results can lead to misdiagnosis and inappropriate therapy decisions. The system must produce consistent and reliable results regardless of variations in input audio quality, speaker characteristics, or recording environments.

The deep learning models are trained on diverse datasets that include multiple speakers, accents, and recording conditions, enabling the system to generalize across different populations. The use of Wav2Vec 2.0 pretrained embeddings provides robust speech representations that have been learned from large-scale speech corpora, improving generalization to unseen speakers and environments.

A confidence threshold mechanism prevents the system from producing unreliable outputs for ambiguous inputs. Predictions falling below the defined threshold are flagged as uncertain rather than being presented as definitive results. Error handling routines capture exceptions that occur due to invalid file formats, corrupted audio, or unexpected input types, preventing complete system failure during operation.

2.3.2 Performance

System performance is essential for making the system practical and usable in real-world clinical environments. The system is designed to deliver analysis results within a few seconds when processing a single audio clip.

This rapid response time is achieved through optimized processing techniques such as lazy loading and caching of models, efficient audio conversion using FFmpeg, and standardizing input to a fixed duration of 3 seconds at 16 kHz. Utilizing pretrained Wav2Vec 2.0 model weights via transfer learning eliminates the considerable resources that would be required to train the model from scratch.

If GPU hardware is available, it is used for accelerated computation during both training and inference. The modular design of the classification and localization pipelines allows parallel processing, improving overall throughput.

2.3.3 Usability

The system is designed to be intuitive and accessible for both clinicians and non-technical users. The user interface provides a clean graphical interface with minimal steps required—simply uploading or recording audio and obtaining results.

Results are presented in plain language that is easy to understand, avoiding technical jargon where possible. Confidence scores accompany all predictions so that users can assess the reliability of the results. For cases where results indicate low confidence, the system suggests consulting a speech-language pathologist.

The system supports both web-based and desktop interfaces, ensuring accessibility across different platforms without requiring specialized hardware or software installation.

2.3.4 Scalability

The system is designed to be scalable from the beginning, with the ability to extend both the size and usage of the system without requiring a complete redesign.

The classification pipeline uses independent binary classifiers, allowing new dysfluency classes to be added by training and incorporating additional models with minimal modifications to the existing codebase. Language-specific adapters enable the system to support additional languages without major architectural changes.

The modular architecture supports horizontal scaling when hosted on cloud platforms, accommodating larger volumes of concurrent users. The separation of preprocessing,

classification, localization, and reporting components allows individual modules to be upgraded or replaced independently.

2.3.5 Maintainability

The codebase follows a modular monorepo architecture, with preprocessing, classification, localization, transcription, and reporting developed as independent but loosely coupled components. This separation of concerns allows developers to modify one section of the system with minimal risk of creating unintended consequences in other sections.

Model files are stored separately and can be replaced with newer versions once retrained on updated data. A centralized model registry with configuration-driven checkpoint management ensures consistent model loading across the web and desktop applications. Unique fingerprinting of model checkpoints enables version tracking and reproducibility.

Extensive documentation and consistent code quality standards, enforced through linting tools, reduce the ramp-up time for new contributors and support ongoing maintenance.

2.4 USER INTERFACE REQUIREMENT

The system provides both a web-based interface and a desktop application to support different user preferences and environments.

2.4.1 Input Page

The input page allows users to interact with the system by providing audio input. Users can upload pre-recorded audio files in WAV, MP3, FLAC, or M4A formats through a clearly marked upload button. Alternatively, users can record speech directly using a microphone via the MediaRecorder API in the web application or the sounddevice library in the desktop application.

The system validates that the uploaded file is in a supported format before proceeding with analysis. A preview of the uploaded audio may be displayed to confirm the correct file has been selected. A clearly visible button initiates the analysis pipeline once the user is ready.

2.4.2 Result Page

The results page displays the output of the analysis pipeline. The classification results indicate which dysfluency types are present in the speech, along with confidence scores for each type. A waveform visualization with highlighted dysfluency regions provides a visual representation of where dysfluencies occur in the audio.

Spectrogram visualizations offer additional insight into the spectral characteristics of the speech signal. A timestamped transcript of the speech is displayed, with detected dysfluencies marked at their corresponding positions. The system also provides an option to generate and download a detailed clinical-style report in PDF format.

2.5 SOFTWARE REQUIREMENT

The software requirements of the proposed system include the tools, frameworks, and technologies used for developing, deploying, and maintaining the application across different platforms.

2.5.1 PyTorch

PyTorch is the primary deep learning framework used for developing and training all machine learning models in the system. It provides the computational backend for Wav2Vec 2.0 feature extraction, binary classification, multitask classification, CNN-based localization, and Wav2Vec2-based localization. PyTorch's dynamic computation graph and automatic differentiation capabilities enable efficient model training and inference. The framework supports GPU acceleration through CUDA and mixed-precision training for improved performance.

2.5.2 Transformers

The Hugging Face Transformers library provides pretrained speech models and integration tools. Wav2Vec 2.0 models (facebook/wav2vec2-base) are used for extracting contextual speech embeddings that serve as input features for classification and localization. The Whisper model is used for automatic speech recognition, generating timestamped transcripts of input audio. Language-specific Whisper variants support English, Kannada, and Hindi transcription.

2.5.3 librosa

librosa is a Python library for audio analysis and processing. It is used throughout the preprocessing pipeline for loading audio files, resampling to the target sample rate, computing mel-spectrograms (128 mel bands, hop length of 512, FFT size of 2048), and performing silence trimming. librosa provides the foundational audio manipulation capabilities that support both training and inference workflows.

2.5.4 Matplotlib

Matplotlib is used for generating visual representations of audio data and model performance. Mel-spectrograms are plotted using Matplotlib for visualization and debugging purposes. Training curves showing accuracy and loss metrics are plotted during model

development to monitor learning progress. Confusion matrices and other evaluation visualizations are generated using Matplotlib to assess model performance across different dysfluency classes.

2.5.5 NumPy

NumPy is used extensively throughout the system for numerical computations on array data. Audio signals are represented as NumPy arrays for preprocessing operations such as DC offset removal, peak normalization, and padding. Model outputs and probability scores are processed using NumPy for aggregation and thresholding. Data preparation and augmentation pipelines rely on NumPy for efficient array manipulation and transformation.

2.5.6 scikit-learn

scikit-learn provides evaluation metrics and tools for assessing model performance. Classification metrics including precision, recall, F1-score, AUROC, and AUPRC are computed using scikit-learn to evaluate the reliability and diagnostic performance of the system. These metrics are used during training to select the best-performing model checkpoints and during evaluation to compare different model architectures and configurations.

2.6 HARDWARE REQUIREMENTS

The hardware requirements define the minimum and recommended system specifications necessary for efficient execution of the proposed system. These requirements ensure smooth performance during both development and deployment phases.

- **Processor:** A system with at least an Intel Core i5 or AMD Ryzen 5 processor (or higher) is required to handle audio processing and model inference efficiently.
- **Memory (RAM):** A minimum of 8 GB RAM is required for basic functionality. However, 16 GB RAM is recommended to ensure smoother performance, especially when handling multiple recordings or running resource-intensive tasks.
- **Storage:** The system requires at least 10 GB of available storage to accommodate datasets, trained model weights, and application files. Additional storage may be needed depending on usage and data accumulation.
- **Graphics Processing Unit (GPU):** An NVIDIA GPU is recommended for training deep learning models, as it enables faster computation through features such as torch.compile, mixed precision, and TensorFloat-32 (TF32). However, a GPU is

not mandatory for inference, and the system can operate on CPU for deployment purposes.

- **Audio Input Device:** A functional microphone is required for recording speech input within the application.

